AFTH Logo

AQUAPRED PROJECT

Smart Monitoring for Thermal Spas

Data-driven quality control and intelligent anomaly detection for balneotherapy.

THE CHALLENGE

The Limits of Traditional Monitoring

- ✕ **Delayed lab results:** Manual sampling is slow.
- ✕ **Reactive protocols:** Fixed rules miss subtle trends.
- ✕ **Hidden dynamics:** Continuous chemical shifts go unnoticed.



European Innovation

Digitizing spa water control across France, Spain, and Portugal.



IoT Sensing

Continuous real-time data acquisition.



Smart Analysis

Predictive modeling and forecasting.



Safe Therapy

Guaranteed optimal mineral benefits.

Real-Time Data Flow



Smart Probes

Physical and chemical sensors.



Cloud Hub

Secure telemetry aggregation.



Intelligent Engine

Advanced predictive analysis.

THE CORE INNOVATION

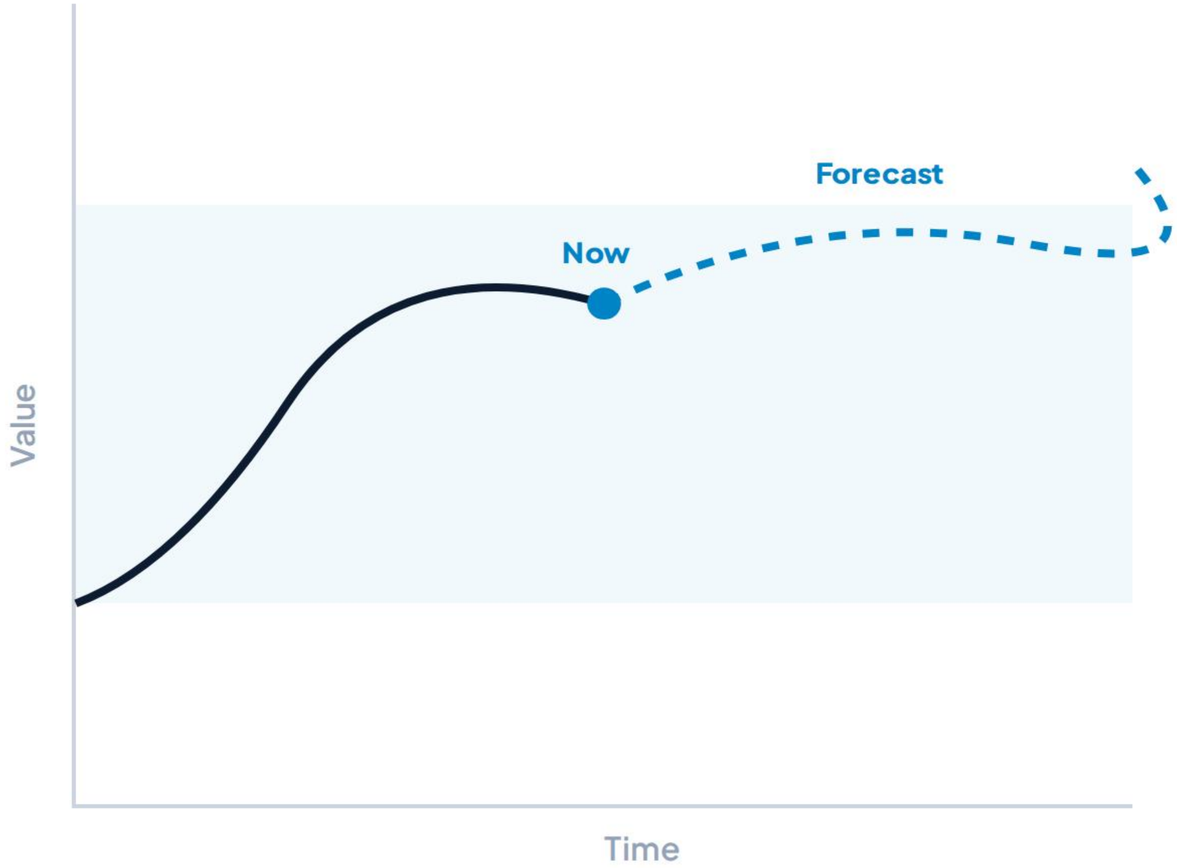
Moving from reactive alarms to proactive intelligence.

USE CASE 01

Predictive Forecasting

Anticipating the short-term evolution of thermal properties.

- Accurate temperature cooling curves.
- Chemical fluctuation anticipation.

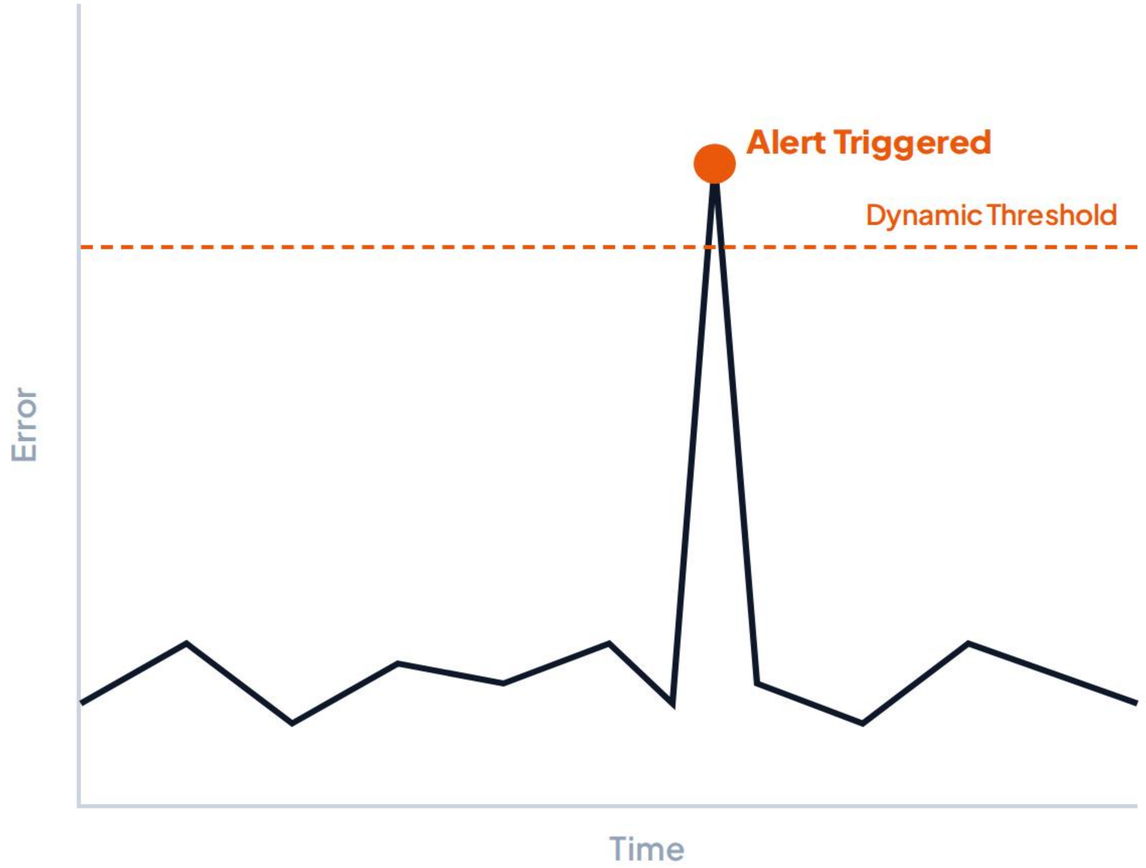


USE CASE 02

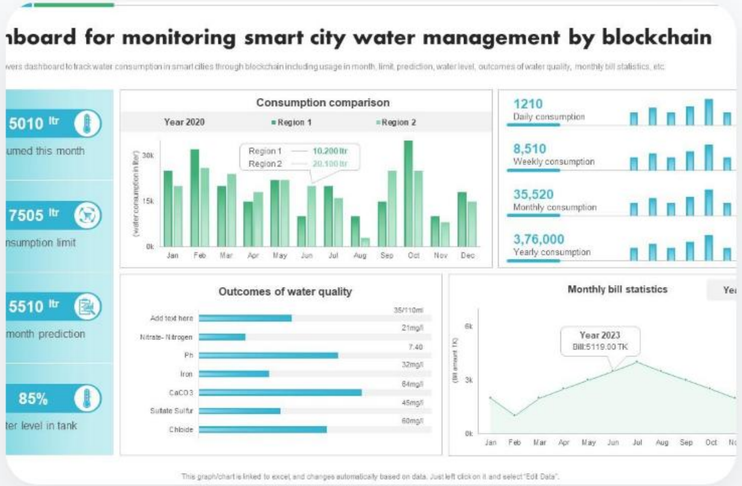
Early Anomaly Detection

Finding hidden issues before they escalate into facility closures.

- Immediate contamination alerts.
- Gradual sensor drift identification.



Beyond Water Quality



Energy Efficiency

Optimized hydraulic heating.



Bather Impact

Occupancy load modeling.

Predictive Maintenance

Automated hardware diagnostics.

Managerial Value

Operational Efficiency

Proactive issue resolution.

Cost Reduction

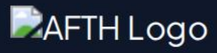
Reduced operational downtime.

Resource Control

Optimized chemical dosing.

Client Assurance

Therapeutic consistency guaranteed.



The Future of Balneotherapy is Digital.

Thank you for your attention.

AQUAPRED PROJECT CONSORTIUM

Image Sources



https://thesanetravel.com/images/NewTravels/ThermesRight/167-poseidon-thesanetravel.com-20230907_161309.JPG

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